

One thing that AI image generation is going to do is suck up ALOT of electricity. How much so I wonder...

Posted by u/zfreakazoidz • 0 points • 14 comments

So when Stable Diffusion came out, I would use it for like 16 hours a day (I'm disabled and got time on my hand). Often the time was just rendering of images. One issue I ran into and why I cut my use down to like an hour a week, my electric bill (in a house of 4 mind you) doubled, if not a bit higher then double.

This does make me wonder what about how much it will effect power grids in the future where AI generation of various types, becomes more popular and main stream. Especially considering we are a "go green" world which means lots of things like cars for example, will also require more drain on the grid.

Assuming of course we don't end up actually figuring out how to make fusion power plants, which produce crazy amonst of electricity.

Comments

u/TheMentalTaco15 • 1 points • 2025-01-18 02:14 UTC

While AI image generation may be no more energy intensive than gaming, the issue is that it's ANOTHER energy intensive thing that is almost objectively less vital to society's culture and economy than gaming. It's not the Satan of sucking up energy, but the issue is that it is another thing we don't need that sucks up energy

u/gerykelf • 4 points • 2024-04-21 18:11 UTC

We are going trough this question the very least once a month.

If we assume you just let the machine rip and push out images in the hundreds (which is a very rare use case), then it is the same as gaming for the same amount of time.

If you are making decisions, inpainting, editing, looking for prompts, etc, then it is a huge spike in GPU usage every couple minutes, and then back to idle. Even less energy consuming than gaming.

Service based AIs such as ChatGPT are eating up a lot of electricity because of the millions using it worldwide for every small thing. If you run them at home even LLMs which are harder to run than image generators, are more power efficient than high end gaming.

Both Stable Diffusion and OogaBooga draws less power and runs cooler on my RTX 3060 12gb than for example Red Dead Redemption 2 with uncapped frames.

And as another example, 3D rendering with blender even surpasses that, as blender is really well designed to use 100% of the computer's resources for faster rendering.

u/EngineerBig1851 • 2 points • 2024-04-21 17:50 UTC

Don't tell this man about primary intent his videocard fulfills...

u/realechelon • 2 points • 2024-04-21 17:36 UTC

My RTX 3090 has a maximum power draw of 350W. It's comparable to a PS5 or a large refrigerator. That's one of the most powerful consumer graphics cards on the market, most use far less energy.

I really don't think that the electric consumption of AI art is going to be a significant factor in the world.

u/88sSSSs88 • 8 points • 2024-04-21 16:08 UTC

A someone correctly said, not much more than gaming on a computer.

Another detail:

Consider how long it takes a digital artist to make a single piece. 1 hour? 10 hours? 20?

Consider how long it takes an AI engine to generate a piece. 8 seconds? 12?

The power draw when AI generating is about 2 or 3x idle.

An AI generated image at 12 seconds per generation consumes about a 100th of the energy that an artist generated image does. And let's be real - most art probably takes considerably more than just an hour.

u/Gimli • 8 points • 2024-04-21 15:25 UTC

Nah. Yeah, it uses the GPU heavily, but you can generate far more than you have the patience to deal with easily. Nobody's going to sit there generating thousands of images for no purpose.

At some point you get bored of picking one image out of a hundred, and start being a bit more careful with the generation.

u/AcceptableLab9729 • -4 points • 2024-04-21 15:22 UTC

I'd rather have blackouts then see anymore shitty furry "art"

u/Economy-Fee5830 • 3 points • 2024-04-21 16:09 UTC

> I'd rather have blackouts then see anymore shitty furry "art"

Do you really NEED to have the blackouts before you see furry "art"? Can you not just view the furry "art" without the blackouts? Is that some kind of foreplay?

u/io_virgil • 2 points • 2024-04-21 15:12 UTC

To calculate the power consumption of the NVIDIA A100 chip over a period of 16 hours, we need to start by understanding the power consumption specifications of the chip under typical operating conditions.

The NVIDIA A100 GPU, designed for data center scale applications, especially in AI and high-performance computing, has several power consumption figures depending on its configuration and use case. Typically, the A100 comes in different power configurations, with a maximum power specification around 250 watts for PCIe card versions and up to 400 watts for the SXM form factor.

To calculate the total power consumed by the NVIDIA A100 over 16 hours, we will assume a continuous operation at its typical maximum power draw. Let's consider the most common scenario where the A100 operates at its full capacity for maximum efficiency:

1. **Power Consumption Rate**: We will take 400 watts as the power consumption rate, which is common in many high-performance settings where the SXM version is used.
2. **Time of Operation**: The total operation time is 16 hours.
3. **Total Power Consumption Calculation**:

Total Power Consumption (kWh) = Power (W) × Time (Hours) / 1000

Plugging in the values:

Total Power Consumption = 400W × 16Hours / 1000 = 6.4kWh

The result of 6.4 kWh represents the energy used by the A100 chip when operated at a peak power of 400 watts over 16 hours. This calculation provides a straightforward estimate of the energy demand for high-performance tasks typical in AI research, deep learning model training, and other compute-intensive operations.

The power consumption of the NVIDIA A100, while significant, is justified by its computational output and efficiency. It supports massive parallel computing tasks and can dramatically reduce the time required for complex calculations compared to less powerful GPUs. However, the high power requirement also necessitates careful consideration of cooling solutions and power supply infrastructure in data centers. Efficient use of such powerful GPUs includes optimizing workload distributions and leveraging power management features to reduce overall energy consumption when full power is not required.

The NVIDIA A100's power consumption profile demonstrates the trade-offs between computational power and energy usage in modern data centers. While the energy cost is considerable, the benefits in terms of performance can outweigh these costs in many applications, especially those requiring intensive computation tasks. The focus on sustainability in computing also pushes for improvements in chip design and system architecture to enhance power efficiency while meeting the growing computational demands.

u/Economy-Fee5830 • 17 points • 2024-04-21 14:36 UTC

It's equivalent to high-end gaming.

So is it really a problem?

u/FakeVoiceOfReason • 1 points • 2024-04-22 17:05 UTC

That's just personal AI models; enterprise models use more.

u/Economy-Fee5830 • 1 points • 2024-04-22 17:16 UTC

Enterprise models will likely be much more efficient than home set-ups, so there is even less to worry about.

u/FakeVoiceOfReason • 1 points • 2024-04-22 17:25 UTC

Well, most professors would probably consider that cheating since one of the points of essay writing is to teach rhetorical and/or research skills, but that's neither here nor there. Arguably, that would only result in energy savings if the person turns off their computer afterwards. And while enterprise models are probably more efficient overall, they're much larger and cost more to run.

u/Economy-Fee5830 • 1 points • 2024-04-22 17:27 UTC

Sure, but usage is task based. A student may turn on their PC for an essay, and then go back to their phone for everything else. A person in a company may need some art for a game background, load up midjourney, and then be done and go home, instead of slaving for 3 days in photoshop making the perfect background art.

Only hobbyists sit in front of computers for fun.

u/FakeVoiceOfReason • 1 points • 2024-04-27 05:55 UTC

Sorry for the delayed response. While usage may be task based, people will generally use their computers for similar amounts of time once they complete a task. Most businesses do eight hour work days (approximately) regardless of how long it takes to complete tasks. If someone finishes early, they're given more tasks. There are certainly exceptions, but this is standard-ish.

Plus, the push to include AI in common workflows (such as Copilot integration on the desktop) increases the cost of those workflows. Unless making computers *more* useful will make people use them *less* overall (which seems unlikely), AI is more likely to increase sum power draw.

That's also ignoring the cost of training models. Training is often considered one-and-done, but many models are being continuously trained as data is generated.

u/Economy-Fee5830 • 1 points • 2024-04-27 09:23 UTC

> Training is often considered one-and-done, but many models are being continuously trained as data is generated.

I don't believe this is true at all, and at scale training is a small part of usage costs.

Regarding using the PC the whole day even when a task is completed, computers have different energy use depending if they are idle vs working hard, so if its just sitting there checking email it uses a lot less energy than if its running photoshop.

Also if people are using copilot to be more productive, this means the company can do more with fewer people, meaning less people employed using energy.

Either way you slice it, getting work done faster and more efficiently saves energy.

u/FakeVoiceOfReason
• 1 points •
2024-04-28
08:08 UTC

Not if training is continuous or near-continuous. If the NPU becomes the standard, presumably it will enable constant learning from a user.

And that's just a local system. Tons of advertising networks are being constantly

trained on live user data.

They certainly do. One of the things we're probably focusing on differently is LLMs vs. SD. If a system like Stable Diffusion *does* enable someone to do more in less time *and* they don't just use that to produce more outputs but actually do something less energy intensive, then it could enable savings (depending on a lot of factors). For LLMs like Copilot, where the output is much less discrete, it's less likely. Document editors typically aren't very resource intensive, so any additional NN processing will likely be additive and unlikely to save much energy. Using ChatGPT instead of Google would probably also take more energy, unless you only use one ChatGPT prompt for several searches.

Unless those people are killed, they're still going to be using energy; just not the company's.

That really is not necessarily true. You can improve speed without improving energy efficiency.

Arguably, the transition to digital art overall improved speed but not energy efficiency.

u/zfreakazoidz • 2 points • 2024-04-21 16:04 UTC

You should more then fine then. I got a i7, 2070 Super (8gb) and do great.

u/Dyeeguy • 3 points • 2024-04-21 14:31 UTC

How will AI impact femboys